

	DOCUMENT TITLE: Equipment Requirements Form	Revision: A
	DOCUMENT #: FM2-QM.SLQ050	Effective Date: 30Jul2022

Equipment Requirements

Equipment ID #:	ST-016	Equipment Description:	Freeze Dry System, 2.5 Liter
Model #:	7670520	S/N:	200594092
Manufacturer:	Labconco Corp.	Operating Range:	Adjustable down to -50°C/-58°F
Type (check one):	☐ Production / ☐ Engineering / ☒ Quality / ☐ Other _____		

Calibration Requirements (check type of calibration):		
☐ Calibration Required Internal / External (circle one) Interval: <u>N/A</u> Internal calibration procedure #: <u>N/A</u>	☐ Limited Calibration For limited calibration, document nature of limited calibration (how it can / cannot be used)	☒ No Calibration For no calibration required, document rationale for reason(s) equipment does not require calibration

Requirements or Rationale for No Calibration:

The Labconco Freeze Dry System is used to freeze dry suspension (C.SLQ001) samples in order to perform the quality control procedure QC-C.SLQ001. The accuracy of the temperature control of the Freeze Dry System does not need to be accurate. It is only required to remove water molecules from the suspension sample. In the event that water molecules are not completely removed from the sample, the water molecules would overwhelm the NMR spectra, and it would be clear that the sample still contained water molecules and that the Freeze Dry System may need repair. In this instance, the suspension sample would be required to undergo additional freeze drying to remove all water molecules. There is no risk to the quality of the suspension sample if the Freeze Dry System is not calibrated.

Preventive Maintenance Requirements (check type of calibration):	
☒ Preventive Maintenance Required Interval: <u>Annual</u> Internal PM procedure #: <u>N/A</u>	☐ No Preventive Maintenance For no preventive maintenance required, document rationale for reason(s) equipment does not require preventive maintenance

Requirements or Rationale for No PM:

Annually:

- 1) Using a vacuum cleaner with brush attachment, clean the refrigeration system condenser to ensure proper airflow for peak performance.
- 2) Check all rubber hoses and gaskets and replace any that show signs of hardening, permanent set or deterioration.
- 3) Using a soft cloth, sponge or chamois and a mild, non-abrasive soap or detergent, clean the acrylic chamber lid.
- 4) Using a soft cloth, sponge or chamois and a mild, non-abrasive soap or detergent, clean the exterior surfaces of the Freeze Dryer. Liquid spray cleaners and polishes may be used on the exterior surfaces. Do not use solvents to remove stains from the exterior surfaces as they may damage the finish.
- 5) Check oil level of the vacuum pump. It should be between MIN and MAX. If the oil level is less than an inch above MIN, add oil to proper level.
- 6) If oil shows cloudiness, particles or discoloration, drain the pump and replace with fresh oil.

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Approval		
	Signature	Date
Quality:	Bryan Wong 	22 Nov 2024
Engineering:	Brian McVerry 	21 NOV 2024